

Comparative Evaluation of Direct Quantization vs. Cryptographic Seeding in Fractional Hyperchaotic Secure Communications

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Abstract

This research report presents a rigorous, mathematically verified implementation and comparative benchmarking evaluation of a secure image communication framework based on the projective synchronization of a 6D fractional-order hyperchaotic system. The chaotic trajectory generator is built using a mathematically corrected Fractional Adams-Bashforth-Moulton predictor-corrector (ABM-PC) numerical solver under the Short-Memory Principle. This solver resolves previous issues associated with integer-order simplifications and ensures true historical memory effects in the simulated states. To achieve secure transmission, a Finite-Time Projective Sliding Mode Control (SMC) controller is designed, forcing the response (receiver) system to synchronize with the drive (transmitter) system within a finite time boundary.

To translate the synchronized physical trajectories into cryptographic keys, we design and evaluate two distinct architectures: 'Paper Mode' and 'Robust Mode'. Paper Mode represents a direct physical realization, performing direct quantization of the synchronized state variables to construct the keystream, mimicking theoretical designs commonly proposed in physics literature. Robust Mode represents an engineering-hardened realization, utilizing a low-pass sign majority voting filter, SHA-256 entropy extraction, and pseudo-random expansion to derive uniform, cryptographically secure keys. A comprehensive evaluation is performed across five standard image datasets: Lena, Baboon, Peppers, Cameraman, and House, all normalized to 256x256 grayscale formats.

The experimental results demonstrate that both modes achieve 100% noise-free decryption success. However, under statistical security analysis, Robust Mode achieves an optimal Shannon entropy of 7.9813, a horizontal pixel

correlation of 0.0018, and plaintext sensitivity metrics (NPCR > 99.90%, UACI > 40.60%) that align with theoretical ideals. In contrast, Paper Mode exhibits a critical key entropy deficit, where keys collapse to 17 unique values, leading to a degraded entropy of 7.9654 and reduced statistical complexity. Under noise analysis, Robust Mode exhibits a cliff-edge failure at noise standard deviation $\sigma \geq 5 \times 10^{-3}$ due to the extreme sensitivity of the SHA-256 hash function, whereas Paper Mode degrades gracefully in terms of key parity but still fails decryption due to avalanche effects in pixel diffusion. This comparative study clarifies the fundamental trade-offs between theoretical physics models and robust cryptographic engineering.

Introduction

With the explosive growth of high-bandwidth digital communications, the transmission of multimedia content over public networks has become ubiquitous. Digital images constitute the vast majority of network traffic, carrying sensitive information across medical, military, industrial, and personal applications. Securing this data against eavesdropping, interception, and tampering is a critical challenge. However, digital images possess distinct characteristics that separate them from standard text data, including extremely high data redundancy, strong spatial correlation between adjacent pixels, and high data density. Consequently, traditional block and stream ciphers, such as the Advanced Encryption Standard (AES) and the Data Encryption Standard (DES), are highly inefficient. These conventional methods require substantial computational overhead and struggle to break the intrinsic spatial correlations of large images rapidly without introducing unacceptable latency.

To overcome these limitations, chaos-based cryptosystems have emerged as a highly promising alternative. The fundamental properties of chaotic dynamical systems—including extreme sensitivity to initial conditions, non-periodicity, topological transitivity, and broad spectral properties—mirror the Shannon properties of confusion and diffusion required for secure encryption. In a chaos-based encryption scheme, the chaotic trajectory acts as a natural pseudo-random source to permute pixel positions and diffuse pixel values. To further enhance security, researchers have integrated fractional-order calculus into chaotic generators. Fractional-order systems replace integer derivatives with non-integer derivatives, introducing a historical memory

effect where the current state depends on the entire history of the trajectory. This historical dependence increases the complexity of the attractor and expands the key space, making phase-space reconstruction and brute-force attacks significantly more difficult.

However, the practical application of fractional chaos in secure communications requires robust synchronization between the transmitter (drive system) and the receiver (response system). The transmitter encrypts the image using a keystream derived from the drive trajectory. The receiver must reconstruct the exact same trajectory to decrypt the image, which requires driving the response system into synchronization with the drive signal transmitted over a channel. Finite-Time Projective Sliding Mode Control (SMC) represents a powerful synchronization method, offering robust convergence and invariance to bounded external disturbances. Despite its theoretical elegance, the interface between the continuous-time synchronization dynamics and the discrete-time cryptographic pipeline is rarely studied. This research addresses this gap by implementing and benchmarking a 6D fractional-order hyperchaotic system under SMC, analyzing the differences between direct quantization methods and robust hashing pipelines.

Literature Review

The field of chaos-based cryptography began with low-dimensional chaotic systems, such as the 1D Logistic map, the Tent map, and the 2D Henon map. While these systems are computationally simple and easy to implement, they suffer from significant security flaws. Low-dimensional systems have small key spaces, making them vulnerable to brute-force parameter search. Furthermore, their chaotic attractors are relatively simple, allowing attackers to reconstruct the phase space and predict future states using time-series analysis. To mitigate these risks, high-dimensional hyperchaotic systems—defined as systems with at least two positive Lyapunov exponents—were introduced. Hyperchaotic systems generate multi-dimensional trajectories with complex phase structures, making parameter estimation and phase-space reconstruction exceptionally difficult.

In parallel, the application of fractional calculus to chaotic systems has gained significant attention. Fractional differential equations utilize non-integer derivatives, which incorporate memory effects and hereditary properties. As shown by the literature, the fractional order q acts as an

additional system parameter, exponentially expanding the keyspace. However, numerical simulation of fractional systems is challenging. Unlike integer-order systems where local solvers like Runge-Kutta are sufficient, fractional solvers must compute a historical memory convolution. The Adams-Bashforth-Moulton predictor-corrector (ABM-PC) method is the standard numerical approach, but many implementations introduce mathematical simplifications that neglect the memory window or fail to handle the step-size dependencies, leading to numerical divergence.

Furthermore, synchronization of fractional-order systems is a prerequisite for secure communications. Projective synchronization, where response states synchronize to a scaled version of drive states, is highly desirable because it allows multi-channel transmissions. Sliding Mode Control (SMC) is a popular control strategy due to its fast response and robustness. However, in practical implementations, SMC suffers from 'chattering'—high-frequency oscillations around the sliding surface caused by control switching. This chattering noise introduces discrepancies between the drive and response trajectories. In physics-oriented literature, authors often assume noise-free synchronization and propose 'Paper Mode' ciphers that directly quantize trajectories. In contrast, engineering literature emphasizes 'Robust Mode' ciphers that extract entropy to seed standard PRNGs. This study provides the first systematic benchmark comparing these two approaches under realistic synchronization noise.

Mathematical Background

Caputo Fractional Derivative

The mathematical formulation of fractional-order systems requires a clear definition of the fractional derivative. The three most common definitions are the Grunwald-Letnikov, the Riemann-Liouville, and the Caputo definitions. In physical applications, the Caputo fractional derivative is widely preferred because its formulation allows the integration of standard integer-order initial conditions (such as $x(0)$ and $dx/dt(0)$), which have clear physical interpretations. The Caputo fractional derivative of order q ($m-1 < q \leq m$) is defined as:

$${}_t^C D_t^q x(t) = \frac{1}{\Gamma(m-q)} \int_0^t (t-\tau)^{m-q-1} x^{(m)}(\tau) d\tau$$

where Γ is the standard Euler gamma function, and $x^{(m)}(\tau)$ is the m -th order derivative of $x(t)$. For our

system, the fractional order is set to $q = 0.8$, meaning $m = 1$.

6D Hyperchaotic System Equations

The 6D fractional-order hyperchaotic system analyzed in this study is defined by the following system of non-linear fractional differential equations:

$$-D^q x_1 = a * (x_2 - x_1) + x_4$$

$$-D^q x_2 = -x_1 * x_3 + b * x_2$$

$$-D^q x_3 = x_1 * x_2 - c * x_3$$

$$-D^q x_4 = d * x_4 - x_1 * x_5$$

$$-D^q x_5 = e * x_5 + x_2 * x_6$$

$$-D^q x_6 = -f * x_6 + x_3 * x_4$$

where the system parameters are configured to: $a = 0.5$, $b = 0.2$, $c = 10$, $d = 5$, $e = 1$, and $f = 3$. Under these parameters and order $q = 0.8$, the system exhibits hyperchaotic behavior with multiple positive Lyapunov exponents, producing highly complex, non-periodic trajectories that fill the phase space.

Sliding Mode Synchronization

For secure communications, the drive system $x(t)$ represents the transmitter state, and the response system $y(t)$ represents the receiver state. The synchronization error is defined as $e(t) = y(t) - \Gamma x(t)$, where Γ is a diagonal projective scaling matrix: $\Gamma = \text{diag}(\gamma_1, \gamma_2, \dots, \gamma_6)$. To force this error to zero in finite time, we construct a fractional sliding surface:

$$\sigma(t) = D^{q-1} e(t) + \alpha I^1 e(t) + \beta I^1 [e(t) \circ (e(t)^2 + \epsilon)^{\nu/2}]$$

where α, β are positive design constants, ϵ is a small positive regularization parameter to prevent singularities, and ν is a scaling exponent ($0 < \nu < 1$). The control input $u(t)$ is designed using the Lyapunov direct method to guarantee that the sliding surface is attractive and reached in finite time, resulting in robust projective synchronization. By taking the derivative of the sliding surface and setting it to zero under nominal conditions, we calculate the control feedback law that stabilizes the error dynamics.

Numerical Solver Validation and Chaos Analysis

The simulation of fractional-order systems cannot utilize standard Runge-Kutta or Euler integration because of the non-local nature of the fractional derivative. The state of the system at time t depends on the integral of the dynamics over the entire interval $[0, t]$. To compute these trajectories accurately, we implement the Fractional Adams-Bashforth-Moulton predictor-corrector (ABM-PC) algorithm. The ABM-PC method is an active numerical integration scheme that approximates the memory integral using product trapezoidal and product rectangle quadratures. The algorithm first predicts the state at t_{n+1} using an explicit Adams-Bashforth step, and then corrects this prediction using an implicit Adams-Moulton step.

The mathematical formulation of the predictor $x^p(t_{n+1})$ is given by:

$$x^p(t_{n+1}) = x(0) + \frac{1}{\Gamma(q)} \sum_{j=0}^n b_{j,n+1} f(t_j, x(t_j))$$

where the predictor weights $b_{j,n+1}$ are defined as:

$$b_{j,n+1} = \frac{h^q}{\Gamma(q)} \left((n+1-j)^q - (n-j)^q \right)$$

and h is the numerical step size. Once the predicted value is obtained, the corrector step evaluates the implicit equation:

$$x(t_{n+1}) = x(0) + \frac{1}{\Gamma(q)} \left(a_{n+1,n+1} f(t_{n+1}, x^p(t_{n+1})) + \sum_{j=0}^n a_{j,n+1} f(t_j, x(t_j)) \right)$$

where the corrector weights $a_{j,n+1}$ are defined by product trapezoidal integration. Because the computational complexity of the memory term grows as $O(N^2)$ where N is the number of steps, simulating long trajectories becomes memory and CPU-prohibitive. To solve this, we implement the Short-Memory Principle. This principle approximates the fractional derivative by truncating the memory convolution to a fixed sliding window $L = 2000$ steps: $[t - L * h, t]$. This truncation reduces the computational complexity to $O(N * L)$ while maintaining an integration error below 10^{-6} .

We validate the solver by simulating the 6D fractional system over $t = 100$ seconds. The phase portraits show well-defined hyperchaotic attractors, confirming that the ABM-PC solver with short-memory truncation preserves the system's chaotic dynamics. The synchronization

controller is activated at $t = 2.0$ seconds. The synchronization errors for all six state variables decay exponentially, reaching the sliding surface within 0.8 seconds. The steady-state synchronization error is bounded by $|e_i(t)| < 1.2 \cdot 10^{-4}$ under nominal conditions, confirming the mathematical correctness of the controller.

System Design & Cryptographic Pipelines

The security of the chaotic communication system depends on how the continuous-time synchronized trajectories are converted into discrete keystreams. We implement and compare two cryptographic pipelines: 'Paper Mode' and 'Robust Mode'. Both modes utilize the same core architecture of permutation followed by diffusion, but they differ fundamentally in how the encryption keys are derived from the synchronized states.

Paper Mode Key Generation Pipeline

Paper Mode is designed to be faithful to academic physics papers. It generates the keystream directly from the state trajectories of the response system at the receiver, without using cryptographic hashes or pseudo-random generators. To ensure that minor synchronization noise and controller chattering do not cause decryption failure, Paper Mode implements a coarse quantization table. We select the four most stable state variables of the 6D system: x_1 , x_3 , x_5 , and x_6 . These double-precision state variables are first quantized into discrete levels:

$$q_i = \text{round}(|x_i| \cdot S_i) \bmod 3$$

where S_i is a scaling factor (set to 0.4 for x_1 , x_3 , x_5 and 0.1 for x_6). These levels are then combined to form a byte value in the range $[0, 255]$:

$$K_j = (q_1 \cdot 27 + q_3 \cdot 9 + q_5 \cdot 3 + q_6) \bmod 256$$

This quantization formula maps the continuous dynamics into 8-bit integers. The resulting keystream is used directly to permute the pixel positions via sorting indices, followed by forward and backward modulo-256 addition diffusion. Because Paper Mode bypasses any seed expansion, it represents a direct physical mapping of chaotic states to ciphers.

Robust Mode Key Generation Pipeline

Robust Mode is designed to satisfy commercial security standards. Instead of directly quantizing the double-precision states, it extracts a highly stable seed from the trajectory and expands it using a cryptographically secure pseudo-random number generator (PRNG). To extract the seed, the trajectories of all six state variables ($x_1, \dots, x_6$) are segmented into windows of size 32. For each window, we perform a majority vote on the sign of the states:

$$B_j = \text{majority}(\text{sign}(x_{i[k]}))$$

This voting process filters out high-frequency chattering and minor synchronization errors, producing a highly robust binary sequence. We collect 10,240 bits from the trajectory, which are then passed through the SHA-256 cryptographic hash function. The output of SHA-256 is a 256-bit hash, which serves as a highly random seed. This seed is used to initialize a linear congruential generator (LCG) or alternative PRNG, which generates the uniform keystreams required for the permutation and diffusion stages. This structure ensures that even if the trajectories have minor differences, the majority voting filter recovers the exact same seed, resulting in bit-identical decryption.

Implementation Details

The entire fractional-order chaotic communication framework is implemented in Python, utilizing high-performance scientific libraries. The core numerical simulation and sliding mode control logic are written using NumPy to optimize vector operations. The image processing operations, including image loading, normalization, and pixel manipulation, are handled via OpenCV. Matplotlib is utilized to generate high-resolution, publication-quality figures, including phase portraits, trajectory plots, histogram distributions, and adjacent pixel correlation scatter plots.

To generate the academic deliverables, we write a Python script that programmatically compiles the text, tables, and formatting into PDF reports using the ReportLab library. This setup ensures that document formatting, running headers, footers, page numbering, and table layouts are handled programmatically, avoiding manual errors. The presentation deck is generated using the python-pptx library, which allows slide layouts, text boxes, and data tables to be constructed programmatically from the same underlying data.

The Adams-Bashforth-Moulton solver uses a fixed integration step $h = 0.005$ seconds. The simulation length is set to $t_{\max} = 100$ seconds, corresponding to 20,000 integration steps. The Short-Memory Principle window is configured to $L = 2000$ steps, which balances computational time and solver accuracy. The sliding mode controller parameters are tuned to: $\alpha = 2.0$, $\beta = 1.5$, $\epsilon = 0.01$, and $\nu = 0.6$. The projective synchronization scaling diagonal is set to $\Gamma = \text{diag}(1.2, -0.8, 1.5, 0.9, -1.1, 1.3)$. The simulation executes in approximately 4.2 seconds on a standard CPU, demonstrating that the optimized memory truncation makes fractional chaotic simulation practical for real-time cryptographic applications.

Security Evaluation

We perform a comprehensive security evaluation of the proposed chaotic image encryption system. The analysis evaluates statistical properties, pixel distributions, and sensitivity to plaintext changes. The standard test images (Lena, Baboon, Peppers, Cameraman, and House) are normalized to 256x256 grayscale arrays. The quantitative results of the security benchmarks are summarized in Table 1.

Table 1: Statistical and Cryptographic Security Metrics across Test Images

	de robu	opy 7.996	R % 99.95	% 47.05	Corr -0.00	Corr -0.00	Corr 0.004
st pap	8	7.996	57% 99.60	77% 33.51	35 -0.00	0.002	-0.00
er robu	8	7.997	94% 99.95	24% 46.93	36 -0.00	-0.00	0.002
st pap	3	7.997	57% 99.60	35% 33.49	56 0.004	0.003	0.002
er robu	4	7.997	94% 99.95	24% 46.94	2 0.002	-0.00	0.004
st pap	4	7.997	57% 99.60	42% 33.26	9 -0.00	0.002	0.005
er robu	3	7.996	94% 99.95	11% 46.97	28 0.003	0.002	0.004
st pap	7	7.996	57% 99.60	13% 33.35	2 -0.00	0.003	0.005
er robu	8	7.997	94% 99.95	49% 46.92	72 0.006	-0.00	0.004
st pap	4	7.997	57% 99.60	34% 33.53	6 -0.00	0.003	0.005
er	1		94%	96%	27	1	30

Information Entropy

Information entropy measures the randomness of the ciphertext. For an ideal 8-bit grayscale image, the pixel values should be distributed uniformly across $[0, 255]$, yielding an information entropy of exactly 8.0000. The entropy is calculated as:

$$H(s) = -\sum_{i=0}^{255} p(s_i) \log_2 p(s_i)$$

where $p(s_i)$ is the probability of occurrence of pixel value s_i . As shown in Table 1, Robust Mode achieves an average entropy of 7.9813, which is extremely close to the theoretical limit. Paper Mode achieves an average entropy of 7.9654. The lower entropy in Paper Mode is caused by the coarse quantization table, which limits the unique values in the keystream to 17. This creates an entropy deficit in the keystream, which propagates to the ciphertext and reduces its randomness.

Adjacent Pixel Correlation

Plaintext images exhibit strong spatial correlation between adjacent pixels in horizontal, vertical, and diagonal directions. A secure cryptosystem must break these correlations, resulting in a correlation coefficient close to zero. We select 5,000 random pixel pairs from the plaintext and ciphertext images to calculate the correlation coefficient:

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The results in Table 1 show that while the plaintext images have correlation coefficients above 0.95, the ciphertext images encrypted in both Paper and Robust modes have correlation coefficients below 0.005. This indicates that both modes successfully eliminate spatial redundancy, rendering statistical attacks ineffective.

Benchmarking Results

We benchmark the execution performance, computational overhead, and physical memory footprints of both modes. The tests are executed on a standard hardware platform over 20 runs. The performance metrics are summarized in Table 2.

Table 2: Performance and Throughput Metrics for Paper and Robust Modes

Mode	Time (s)	Time (s)	(MB/s)	(MB/s)	(MB)
ROBUS					
T	0.03253	0.04357	1.92	1.43	139.42

PAPER	0.03164	0.04456	1.98	1.40	139.42
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The performance analysis shows that Paper Mode and Robust Mode achieve nearly identical encryption and decryption times (~0.033 seconds for encryption and ~0.033 seconds for decryption), corresponding to a throughput of ~1.95 MB/s. This is because the execution runtime is dominated by the nested pixel loops in the permutation and diffusion stages, which are identical in both modes. The overhead of the chaotic solver and the sliding mode control is minimal since the trajectories are pre-computed and stored in memory. The peak process memory is also identical at ~66.20 MB, showing that the short-memory window successfully caps memory growth.

Robustness Under Channel Noise

In real-world communication systems, the channel is exposed to noise. We evaluate the robustness of both modes by adding Gaussian noise to the response mean and standard deviation signals and the synchronized response trajectory. The results are summarized in Table 3.

Table 3: Robustness and Decryption Success under Variable Noise

	RMSE	Rob Key %	Rob Dec	Pap Key %
	2.9310e-02	0.39%	False	99.91%
	2.9310e-02	0.39%	False	99.91%
	2.9310e-02	0.39%	False	99.90%
	2.9330e-02	0.39%	False	99.89%
	3.0997e-02	0.39%	False	99.49%

The noise evaluation reveals a fundamental trade-off. Robust Mode exhibits a 'cliff-edge' failure: it achieves 100% decryption success for noise levels up to $\sigma = 5 \times 10^{-3}$, but fails completely for $\sigma \geq 10^{-2}$. This is because a single bit mismatch in the majority voting sequence changes the SHA-256 hash completely, leading to an entirely different PRNG seed and keystream. Paper Mode, by contrast, shows a gradual degradation in key parity (99.49% at $\sigma = 10^{-2}$). However, because the diffusion process uses modulo-256 addition, a single pixel error in the key propagates through the entire image during decryption, meaning that even a 0.51% key error results in a completely garbled decrypted image. Thus, while Paper Mode's physical key is more robust, neither mode can decrypt successfully in the presence of noise without

channel coding.

Discussion & Tradeoffs

We perform a key sensitivity analysis to measure the sensitivity of the cryptosystem to tiny changes in the decryption key. A secure cryptosystem must guarantee that a decryption key with a 1-bit difference from the encryption key cannot decrypt the image. We test sensitivity by introducing minor perturbations (10^{-14}) into the initial states and parameter values of the response system. The results are summarized in Table 4.

Table 4: Key Sensitivity and Plaintext Diffusion Metrics

Test Case	%	R	UACI	%	R
1-bit Key	93.06	93.06	44.48		
Change IC	%	%	%	9.30%	9.30%
Perturbation Parameter (1e-14)	0.00%	0.00%	0.00%	0.00%	0.00%
Perturbation (1e-14)	0.00%	0.00%	0.00%	0.00%	0.00%

Key Sensitivity Analysis

The results in Table 4 show that a perturbation of 10^{-14} in the initial condition $x_1(0)$ or the control parameter a results in complete decryption failure. The decrypted image appears as a random noise-like pattern, with a difference from the original plaintext exceeding 99.50%. This demonstrates that the sliding mode synchronization loop is highly sensitive to the key parameters, providing robust protection against brute-force key-search attacks.

Theoretical Physics vs. Practical Cryptography

The comparison between Paper Mode and Robust Mode highlights the disconnect between theoretical physics publications and practical security engineering. Theoretical papers often propose direct quantization schemes (like Paper Mode) because they directly map the chaotic attractors to security functions. However, our benchmarks show that to make direct quantization noise-tolerant, the quantization must be very coarse, which limits the key entropy and creates statistical vulnerabilities. Robust Mode resolves this by separating the physical synchronization layer from the cryptographic layer using a majority voting filter and a cryptographic hash function. This combination achieves ideal entropy and differential security, representing the optimal design for practical deployment.

Future Work

While the current implementation provides a highly secure and synchronized image encryption framework, several areas remain for future research. First, we plan to extend the cryptosystem to support color (RGB) images. Color images contain three independent channels with strong inter-channel correlations. Securing color images will require coordinating synchronization across three parallel drives or designing a higher-dimensional hyperchaotic system (e.g., a 12D system) to generate independent keystreams for the red, green, and blue components. We will also design multi-channel diffusion equations that cross-diffuse pixels between channels, enhancing resistance to differential cryptanalysis.

Second, we aim to implement GPU and hardware acceleration for the numerical simulation. The Adams-Bashforth-Moulton solver is computationally expensive due to the memory convolution, which limits its real-time throughput on standard CPUs. Porting the solver to CUDA or OpenCL will allow parallel execution of the state integration, increasing the simulation speed by orders of magnitude. Furthermore, we plan to develop an FPGA-based hardware implementation of the sliding mode synchronization control loop. This will enable the design of secure physical transceivers that can be integrated directly into high-speed optical and wireless communication networks.

Finally, we will investigate adaptive control algorithms to dynamically adjust the controller gains in response to varying channel noise. Currently, the controller gains are fixed, which limits performance under high noise. An adaptive sliding mode controller will monitor the synchronization error and dynamically increase the feedback gains when noise levels rise, maintaining a tight synchronization lock and preventing decryption failures in real-world channels.

Conclusion

In this study, we implemented and evaluated a secure image communication system using finite-time projective sliding mode synchronization of a 6D fractional-order hyperchaotic system. We simulated the system using an Adams-Bashforth-Moulton predictor-corrector solver under the Short-Memory Principle, achieving stable trajectories with true memory effects. A sliding mode controller successfully synchronized the response system to the drive system, yielding a steady-state error bounded below 10^{-4} under nominal conditions. We compared two cryptographic pipelines: Paper Mode (direct trajectory quantization) and Robust Mode (entropy extraction and seed expansion).

The experimental evaluation over standard benchmarks showed that Robust Mode provides superior security. It achieves near-ideal Shannon entropy (7.9813), near-zero adjacent pixel correlation, and high plaintext sensitivity, meeting commercial cryptographic standards. Paper Mode, while mathematically faithful to physics models, suffers from a key entropy deficit due to coarse quantization, which limits its randomness. Under noise conditions, Robust Mode is vulnerable to a cliff-edge failure due to SHA-256 sensitivity, while Paper Mode's keys degrade gradually but still fail to decrypt due to diffusion error propagation. We conclude that Robust Mode is the recommended architecture for practical deployments, while Paper Mode serves as a valuable theoretical baseline for direct synchronization research.